

Egor Petrov

PhD Student, Computer Science, Columbia University

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Research interests: efficient large-scale LLM training – pretraining & post-training, parallelisms, optimization, scaling laws.

Education

Columbia University – PhD in Computer Science

New York, NY | 09/26 – 05/31 (expected)

SEAS Graduate Student Fellowship.

Moscow Institute of Physics and Technology (MIPT)

Moscow | 2022 – 2026

B.S. in Applied Mathematics and Physics (graduated 06/26). Data Science Track (MIPT & Yandex School of Data Analysis): ML, DL, RL, CV, NLP, RecSys, Bayesian Statistics, Time Series.

Research & Engineering Experience

DAP Lab, Columbia University – PhD Researcher, Post-Training Efficiency New York | Sep 2026 – present

- Studying **optimization dynamics in RL post-training (RLVR)**: benchmarking optimizer behavior and loss-landscape changes relative to pretraining, and their effect on downstream tasks.
- Developing **staleness-mitigation methods for asynchronous post-training** with parallel rollout generation, co-designing algorithms and systems to maximize GPU throughput.

Yandex Research – ML Researcher, Pretraining

Moscow | Jul 2025 – Aug 2026

- Led research on asynchronous **pipeline parallelism** for efficient large-scale LLM pretraining; first-author paper accepted to **ICML 2026**.
- Designed and analyzed novel algorithms for **distributed training systems**, improving scalability and hardware utilization.

Basic Research in AI Laboratory (BRAIn Lab) – ML Researcher

Moscow | Feb 2024 – Aug 2026

- Pioneered memory-efficient zeroth-order optimization methods for LLM fine-tuning, achieving a **50% reduction in memory footprint** (ICML Workshop 2025).
- Developed and theoretically analyzed novel stochastic and zeroth-order algorithms for decentralized optimization, validated across multiple domains (UAI 2025, ICLR 2026).
- Derived a full **second-order (Hessian) characterization** of the Transformer block with implications for scaling laws.

Yandex – ML Engineer, Personalization Quality R&D Group

Moscow | Aug 2024 – Jan 2025

- Engineered a production pipeline integrating Vowpal Wabbit-based online features into a large-scale CatBoost model, achieving a **0.1% uplift in core ranking metrics** (AUC, nDCG).
- Implemented a parallelized, MapReduce-style data converter that **accelerated data processing by 20x**, reducing model training time from hours to minutes.

Publications (* equal contribution)

- One-Step Gradient Delay is Not a Barrier for Large-Scale Asynchronous Pipeline Parallel LLM Pretraining** – P. Zmushko*, **E. Petrov***, N. Abdullaev, M. Khrushchev, S. Horváth. *ICML 2026*. [\[Paper\]](#)
- Sign-SGD is the Golden Gate between Multi-Node and Single-Node Learning** – D. Medyakov, S. Stanko, G. Molodtsov, P. Zmushko, G. Evseev, **E. Petrov**, et al. *ICLR 2026*. [\[Paper\]](#)
- Leveraging Coordinate Momentum in SignSGD and Muon: Memory-Optimized Zero-Order LLM Fine-Tuning** – **E. Petrov**, G. Evseev, A. Antonov, A. Veprikov, P. Plyusnin, N. Bushkov, et al. *ICML Workshop on Tiny Titans, 2025*. [\[Paper\]](#)
- When Extragradient Meets PAGE: Bridging Two Giants to Boost Variational Inequalities** – G. Molodtsov, V. Parfenov, **E. Petrov**, E. Grigoriy, D. Medyakov, A. Beznosikov. *UAI 2025*. [\[Paper\]](#)
- Shuffling Heuristic in Variational Inequalities: Establishing New Convergence Guarantees** – D. Medyakov, G. Molodtsov, E. Grigoriy, **E. Petrov**, A. Beznosikov. *ICOMP 2024*; submitted to JOTA (Q1). **Best student paper**, Neuroinformatics 2024. [\[Paper\]](#)
- Sampling of Semi-Orthogonal Matrices for the Muon Algorithm** – **E. Petrov**, G. Evseev, A. Antonov, A. Veprikov, N. Bushkov, S. Moiseev, A. Beznosikov. *AI Journey 2025*; published in Doklady Mathematics. [\[Paper\]](#)
- Zero Order Algorithm for Decentralized Optimization Problems** – A. Veprikov, **E. Petrov**, G. Evseev, A. Beznosikov. *AI Journey 2024*; published in Doklady Mathematics. [\[Paper\]](#)
- Closing the Curvature Gap: Full Transformer Hessians and Their Implications for Scaling Laws** – **E. Petrov**, N. Kiselev, V. Meshkov, A. Grabovoy. Submitted to *NeurIPS 2026*. [\[Paper\]](#)

Working Papers & Ongoing Projects

- Rethinking Memory-Efficient Optimizers for LLM Pre-Training.** Systematic study of structured optimizer-state compression vs. state quantization under matched memory budgets; peak-memory and communication analysis of optimizer states under sharding (FSDP-style). *In submission to ICLR 2027*.
- Hyperparameter Transfer and Scaling Laws for Asynchronous Pretraining.** Theoretical and practical transfer rules (learning rate, momentum, batch size, weight decay) and scaling laws in models and tokens for asynchronous pipeline-parallel training; extends our ICML 2026 results. *Experiments in progress*.
- Optimization in RL Post-Training (RLVR).** Benchmarking optimizer behavior, loss-landscape changes, and

downstream-task effects when moving from pretraining to RL post-training. *Early stage*.

- **Asynchronous Post-Training.** Mitigating policy staleness from parallel rollout generation; algorithm–systems co-design for maximizing GPU throughput in RL post-training. *Early stage*.

Open Source & Invited Talks

- **ZO-Library, Lead Developer** — open-source library of state-of-the-art zeroth-order optimization algorithms with a `torch.optim`-style API. [\[GitHub\]](#)
- **Aboba, Maintainer** — Python library for A/B testing: statistical tests plus CUPED and linearization via an intuitive API. [\[Webpage\]](#)
- **Talks:** *Large-Scale Training @ ICML 2026* (Yandex School of Data Analysis, 07/2026); *Memory-Efficient LLM Training* (BRAIn Lab Open Reading Group, 11/2025); *SimCLR: Overview and Implications* (Kharkevich Institute, RAS, 06/2024).

Teaching & Selected Awards

- **Teaching Assistant, Central University** (2025–2026): Introduction to AI, Scientific Studio, Python for Data Analysis; mentored student research projects. **Teaching Assistant, MIPT** (2024–2025): Algorithms and Data Structures.
- **SEAS Graduate Student Fellowship**, Columbia University (2026); **B.T. Polyak Scholarship** for contributions to numerical optimization methods (2025); **Best Student Paper**, Neuroinformatics 2024; **Increased State Academic Scholarship**, MIPT (2024–2026).

Technical Skills

- **Programming:** Python, C++, SQL **ML/DL:** PyTorch, Transformers (HF), NumPy, Pandas, Scikit-learn, TensorFlow
- **Distributed training:** data / tensor / pipeline / sequence / expert parallelism, FSDP-style sharding, memory-efficient optimizers
- **Tools & Infrastructure:** Git, Docker, Apache Hadoop, Kafka, Airflow, Vowpal Wabbit, CMake **Languages:** Russian (native), English (fluent; TOEFL 102)